Lesson Planning

Complete Lesson Plan Book

Class: Class 8

Subject: Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Atoms and Molecules

Learning Objectives

• Define atom and molecule.

• State Dalton's atomic theory in simple words.

• Distinguish between elements, compounds and mixtures at particle level.

• Write chemical formulae of simple compounds.

Key Concepts

• An atom is the smallest particle of an element.

• A molecule is two or more atoms joined together.

• Atoms of the same element are alike; different elements have different atoms.

• Water (H_2O) and carbon dioxide (CO_2) are compounds.

• Subscripts in formulae show the number of each atom.

Teaching Methodology (Step by Step)

1. Show models: single balls (atoms) joined in pairs (molecules).
2. Explain Dalton's theory with four simple points.
3. Classify pictures into atoms, molecules, compounds and mixtures.
4. Teach writing formulae: cross the valencies.
5. Students build molecules with ball-and-stick models.

Classroom Activities

• Build H_2O, O_2 and CO_2 with modelling clay and sticks.

• Formula-writing race in teams.

Assessment Questions

1. Define atom and molecule.
2. Write the formulae of sodium chloride and calcium oxide.
3. Is air an element, compound or mixture? Why?

Homework

• Write the formulae of ten common compounds.

Time Allocation

Models 5 min, teaching 20 min, building 10 min, quiz 5 min.

Unit 2: Chemical Reactions

Learning Objectives

- Define a chemical reaction.
- Identify signs of a chemical reaction.
- Write and balance simple chemical equations.
- Distinguish between combination and decomposition reactions.

Key Concepts

- Reactants change into products in a chemical reaction.
- Signs: colour change, gas bubbles, heat, precipitate, smell.
- A word equation names reactants and products.
- Balanced equations have equal atoms on both sides.
- Combination joins substances; decomposition breaks them apart.

Teaching Methodology (Step by Step)

1. Demonstrate: vinegar + baking soda fizzing.
2. List the five signs of reaction with examples.
3. Write the word equation for the demo.
4. Balance $H_2 + O_2 \rightarrow H_2O$ step by step.
5. Students balance three given equations.

Classroom Activities

- Safe demo observations in the lab.
- Balance-the-equation card game.

Assessment Questions

1. Name four signs of a chemical reaction.
2. Balance: $Mg + O_2 \rightarrow MgO$.
3. Give one example each of combination and decomposition.

Homework

- Write word and balanced equations for three reactions.

Time Allocation

Demo 10 min, teaching 20 min, practice 10 min.

Unit 3: Light: Reflection and Refraction

Learning Objectives

• State that light travels in straight lines.

• State the laws of reflection.

• Distinguish between real and virtual images.

• Explain refraction with daily-life examples.

Key Concepts

• Light travels at 300,000 km/s in a straight line.

• Angle of incidence equals angle of reflection.

• A plane mirror forms a virtual, upright image.

• Refraction is the bending of light at a boundary.

• A straw in water looks bent because of refraction.

Teaching Methodology (Step by Step)

1. Shine a torch through smoke to show a straight beam.
2. Demonstrate reflection with a mirror and ray box.
3. Measure angles to verify the laws of reflection.
4. Show the bent-straw and coin-in-water refraction demos.
5. Students draw ray diagrams for a plane mirror.

Classroom Activities

• Periscope making with two mirrors.

• Refraction observation journal.

Assessment Questions

1. State the two laws of reflection.
2. Why does a straw look bent in water?
3. Draw the ray diagram for a plane mirror image.

Homework

• List five examples of reflection and refraction at home.

Time Allocation

Demos 10 min, teaching 20 min, diagrams 10 min.

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Unit 4: Sound

Learning Objectives

• Explain that sound is produced by vibrations.

• State that sound needs a medium to travel.

• Distinguish between loudness and pitch.

• Explain echoes and give uses of sound reflection.

Key Concepts

• Vibrating objects produce sound.

• Sound travels through solids, liquids and gases, but not vacuum.

• Loudness depends on amplitude; pitch depends on frequency.

• An echo is reflected sound.

• Bats and ships use echoes to locate objects.

Teaching Methodology (Step by Step)

1. Strike a tuning fork and touch it to water to show vibrations.
2. Bell-in-vacuum-jar thought experiment for the medium.
3. Demonstrate loud vs soft and high vs low pitch.
4. Clap in the hall to hear an echo.
5. Students measure the speed idea with a simple echo timing.

Classroom Activities

• Make string telephones in pairs.

• Instrument making with bottles of water.

Assessment Questions

1. How is sound produced?
2. Why can't sound travel in a vacuum?
3. Distinguish between loudness and pitch.

Homework

• Write five uses of sound reflection.

Time Allocation

Demos 10 min, teaching 20 min, making 10 min.

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Unit 5: Electricity: Current and Circuits

Learning Objectives

• Define electric current and its unit.

• Draw simple circuit diagrams.

• Distinguish between series and parallel circuits.

• State safety rules for electricity.

Key Concepts

• Current is the flow of electric charge, measured in amperes.

• A circuit needs a source, wires and a load.

• Series: one path; if one bulb fuses, all go off.

• Parallel: separate paths; other bulbs stay on.

• Never touch switches with wet hands.

Teaching Methodology (Step by Step)

1. Show a torch cell, bulb and wires; make it light.
2. Introduce circuit symbols and draw a simple diagram.
3. Build series and parallel circuits with batteries and bulbs.
4. Compare brightness and behaviour when one bulb is removed.
5. Teach five electrical safety rules.

Classroom Activities

• Build series and parallel circuits in groups.

• Design a safety poster for home electricity.

Assessment Questions

1. Draw a circuit with a cell, switch and two bulbs in series.
2. What happens in a parallel circuit if one bulb fuses?
3. Write three electrical safety rules.

Homework

• Draw your home's wiring plan showing series/parallel use.

Time Allocation

Demo 10 min, teaching 15 min, building 10 min, safety 5 min.

Unit 6: Plant Systems: Transport and Reproduction

Learning Objectives

• Explain how water moves up in plants.

• Describe photosynthesis in simple words.

• Label the parts of a flower.

• Explain pollination and fertilisation briefly.

Key Concepts

• Xylem carries water up; phloem carries food.

• Photosynthesis: carbon dioxide + water → glucose + oxygen (in sunlight).

• Flower parts: sepals, petals, stamens, carpel.

• Pollination is the transfer of pollen to the stigma.

• After fertilisation, the ovary becomes the fruit.

Teaching Methodology (Step by Step)

1. Place a white flower in coloured water to show transport.
2. Revise the photosynthesis equation with a leaf demo.
3. Dissect a flower and label its parts.
4. Explain pollination with bee pictures.
5. Students label a flower diagram.

Classroom Activities

• Flower dissection in the lab.

• Coloured-water transport observation over two days.

Assessment Questions

1. Name the tissues that transport water and food in plants.
2. Label the parts of a flower.
3. What is pollination?

Homework

• Draw a flower and label its reproductive parts.

Time Allocation

Demo 10 min, teaching 20 min, labelling 10 min.

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Unit 7: Reproduction in Animals and Plants

Learning Objectives

• Distinguish between sexual and asexual reproduction.

• Describe the human reproductive system briefly and respectfully.

• Explain vegetative propagation in plants.

• State the importance of reproduction for species survival.

Key Concepts

• Sexual reproduction needs two parents; asexual needs one.

• In humans, fertilisation forms a zygote that grows in the uterus.

• Plants reproduce asexually by runners, bulbs and cuttings.

• Seeds are the product of sexual reproduction in flowering plants.

• Reproduction ensures the continuity of life.

Teaching Methodology (Step by Step)

1. Discuss: how do potato plants grow from eyes?
2. Explain asexual reproduction with plant examples.
3. Introduce sexual reproduction in plants: flower to seed.
4. Present the human system with scientific, respectful language.
5. Emphasise respect, privacy and asking trusted adults questions.

Classroom Activities

• Grow a plant from a cutting and record growth.

• Sort picture cards into sexual and asexual reproduction.

Assessment Questions

1. Distinguish between sexual and asexual reproduction.
2. Name two methods of vegetative propagation.
3. Why is reproduction important?

Homework

• Write five examples of asexual reproduction in plants.

Time Allocation

Discussion 5 min, teaching 25 min, activity 10 min.

Unit 8: The Solar System and Beyond

Learning Objectives

• Name the planets and state one fact about each.

• Explain the phases of the Moon.

• Distinguish between stars, planets and satellites.

• State Pakistan's achievements in space (SUPARCO).

Key Concepts

• The Sun contains 99.8% of the solar system's mass.

• Moon phases are caused by the changing sunlit portion we see.

• A satellite orbits a planet; the Moon is Earth's natural satellite.

• Stars are huge balls of hot gas.

• SUPARCO is Pakistan's space agency.

Teaching Methodology (Step by Step)

1. Revise planet order with the class 7 memory sentence.
2. Use a lamp and ball to demonstrate Moon phases.
3. Compare star, planet and satellite with a table.
4. Share SUPARCO's satellites and achievements.
5. Students create a phases-of-the-Moon flip chart.

Classroom Activities

• Moon-phase flip book.

• Research: one space mission and present it.

Assessment Questions

1. What causes the phases of the Moon?
2. Distinguish between a planet and a satellite.
3. What is SUPARCO?

Homework

• Observe the Moon for a week and sketch its phases.

Time Allocation

Demo 10 min, teaching 20 min, flip book 10 min.

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